SQL for Data Analysts

Module 01: Database Fundamentals Handbook

A Comprehensive, Mobile-Friendly Travel Reading Guide

Prepared for Your Data Analyst Journey

Module 01: Database Fundamentals

1. Introduction to SQL

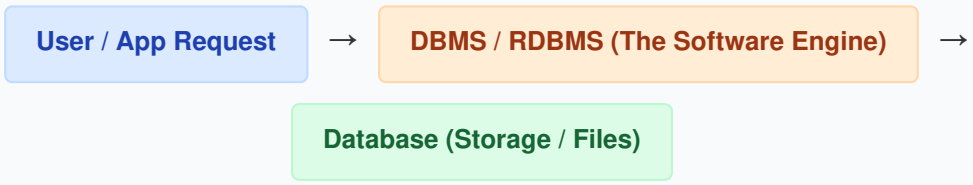
SQL (Structured Query Language) is the universal domain-specific language used to manage, query, and manipulate data stored in relational databases. For a Data Analyst, SQL is the single most critical tool. While tools like Excel handle small datasets, SQL effortlessly handles millions of rows of data across corporate servers.

Think of SQL as your conversation tool with data. Instead of looking for patterns manually, you write concise SQL code to extract high-value insights, track revenue changes, or profile customer behavior instantly.

2. What is a Database, DBMS, and RDBMS?

To understand the structural backbone of modern tech companies, we must differentiate between where data is kept, how it is managed, and how it forms relationships.

Figure 1: The Core Database Architecture



- **Database:** A secure physical container on a drive where raw records reside. Without software, it is just a static warehouse of data.
- **DBMS (Database Management System):** The systematic software layer operating on top of a database. It handles user access, file organization, modification requests, and storage allocation. Examples include classic file-management utilities.
- **RDBMS (Relational Database Management System):** An advanced DBMS built around the concept of organized, cross-referenced grids called **tables**. Every table is interconnected based on shared information, ensuring zero structural clutter and perfect structural mapping.

3. SQL vs. MySQL

This is a foundational point of confusion for beginners. They are entirely separate entities with distinct roles:

- **SQL is a Standardized Language:** It represents the grammar rules, verbs, syntax guidelines, and text keywords used to articulate data operations. It is governed internationally by ANSI standards.
- **MySQL is a Software Application:** It is a specific database product created by Oracle that reads your SQL instructions and executes them on hard drives. Alternate software engines that read SQL include PostgreSQL, Microsoft SQL Server, and Oracle Database.

4. Tables, Rows, and Columns

Within any relational system, data is structured intuitively into rows and columns, heavily mirroring an advanced, programmatic version of an Excel spreadsheet.

Figure 2: Structure of a Database Table ("customers")

customer_id (Column/Field)	first_name (Column/Field)	city (Column/Field)
101 *(Row 1 / Record)*	Amit	Mumbai
102 *(Row 2 / Record)*	Priya	Bengaluru

- **Table:** A unique, named dataset dealing with one concept (e.g., users, transactions, or stock inventory).
- **Columns (Fields):** Vertical divisions defining *what structural trait* is tracked. Every single block under a column shares the same Data Type (e.g., numbers, text strings, or dates).
- **Rows (Records / Tuples):** Horizontal collections containing a single, unique entity's parameters (e.g., everything about customer Amit).

5. Primary Keys and Foreign Keys

To avoid massive redundancy and ensure distinct identification, databases utilize structural anchors known as **Keys** to establish logical relationships.

Figure 3: Key Connections Between Tables

Table: customers

• **customer_id [Primary Key]** | first_name | city

• 101 | Amit | Mumbai

Table: orders

• order_id [Primary Key] | **customer_id [Foreign Key]** | amount

• 5001 | 101 | 45000

- **Primary Key (PK):** A dedicated column guaranteeing that every row has a distinct identity. A Primary Key column **cannot be empty (NOT NULL)** and **cannot contain duplicates**. Example: Student Roll IDs or unique Passport numbers.
- **Foreign Key (FK):** A tracking column in one table that mirrors the Primary Key of another table. It acts as an relational link pointing from one file back to another, ensuring you can map data across the system without replicating entire sheets.

6. Types of SQL Commands (DDL, DML, DQL, DCL, TCL)

SQL syntax commands are broken up into 5 categorical groupings depending on their structural objective:

Category	Full Name & Purpose	Primary Commands Used
DQL	**Data Query Language:** Used to fetch, read, and extract reports from existing datasets. The vital core area for Data Analysts.	`SELECT`
DML	**Data Manipulation Language:** Used to edit, insert, change, or erase actual row-level text and numerical records.	`INSERT`, `UPDATE`, `DELETE`
DDL	**Data Definition Language:** Used to map out, change, or remove the core structural blueprints (tables and fields).	`CREATE`, `ALTER`, `DROP`
TCL	**Transaction Control Language:** Controls how operations are structurally saved or safely undone to maintain system health.	`COMMIT`, `ROLLBACK`
DCL	**Data Control Language:** Manages internal system user permissions, login access parameters, and security restrictions.	`GRANT`, `REVOKE`